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Evaluating AI-assisted creative ideation: A crossover study in higher education

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ABSTRACT

As artificial intelligence (AI) becomes increasingly integrated into educational contexts, its impact on students' creative thinking remains unclear. Given the critical role of creativity in engineering and design education, understanding how AI tools shape students' ideation processes is essential for developing effective pedagogical practices. This study examines the impact of generative AI, specifically ChatGPT, on creative ideation among undergraduate design engineering students. The research was conducted through a randomised crossover experiment, with students alternating between AI-assisted and unaided ideation tasks. A mixed-methods approach combined quantitative analyses of the 728 ideas generated with a qualitative evaluation of students' interactions with AI. Results show that the use of AI did not reduce fluency, flexibility, or originality, nor did it lead to thematic homogenisation. However, semantic divergence was significantly lower in the AI-assisted condition, suggesting convergence in the way ideas were formulated. Additionally, AI-assisted ideas more frequently resembled existing products and exhibited reduced textual elaboration. A mixed between-within subjects ANOVA revealed that students who began with AI support produced more original and diverse ideas across both tasks, pointing to a lasting effect of early AI use. Qualitative analysis of student-AI interactions revealed important patterns, including predominantly passive and directive use, with limited exploratory or collaborative engagement. These findings provide new insights into human-AI co-creation and highlight the importance of promoting intentional, critical, and pedagogically guided use of generative AI tools in education.

1. Introduction

As global challenges such as climate change, economic inequality, and technological disruption call for innovative and interdisciplinary solutions, fostering students' creative capacities has become imperative (Henriksen et al., 2018; Richardson & Mishra, 2018). International organisations, including UNESCO (2016) and the United Nations (2015), have emphasised the importance of creativity in advancing sustainable development, identifying it as a key competency that educational institutions must cultivate. In response, universities should create environments that nurture imagination, experimentation, and problem-solving. This emphasis is particularly critical in design and engineering education, where creative thinking is not a complementary skill but a foundational one, essential for identifying complex problems and envisioning adaptive, forward-thinking solutions (Badran, 2007; Chen & Hsu, 2006; Cropley, 2020).

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Contemporary research defines creativity as a dynamic process that can be learned and shaped by social interactions, tools, and collaborative environments rather than a fixed individual trait (Foglia & Wilson, 2013; Glăveanu, 2010; Sawyer & DeZutter, 2009; Stanciu, 2015). This view advocates for educational strategies that help students develop divergent thinking and encourage creative exploration within supportive learning contexts.

In this context, the emergence of artificial intelligence (AI) technologies has introduced new challenges and possibilities, prompting critical reflection on their implications for creativity and learning. Early studies suggest AI may enhance creative processes (Habib et al., 2024; Hubert et al., 2024; Li et al., 2024; Wieland et al., 2022). However, ongoing debates question whether it can genuinely participate in creative thought, given its lack of emotional expression, contextual understanding, and intentionality (Erden, 2010; Moruzzi, 2025; Wingström et al., 2022). Critical gaps remain in understanding how students engage with these tools in real learning contexts. In particular, little is known about how the timing of AI exposure affects creative processes, its potential to homogenise ideas, and how students interact with these tools during ideation (Ahmad et al., 2023; Anderson et al., 2024; Chandrasekera et al., 2024; Marrone et al., 2022; Moon et al., 2024).

This research addresses these gaps by examining how generative AI shapes ideation processes within classroom environments. Specifically, using a randomised controlled crossover experiment and a mixed-methods approach, this study investigates the integration of generative AI into academic ideation tasks with four main objectives:

1. To evaluate the impact of AI use on creativity outcomes.
2. To examine how the order of AI exposure influences students' performance on subsequent creative tasks.
3. To assess whether the use of AI contributes to the homogenisation of creative outputs.
4. To explore the interaction patterns that emerge between students and AI during the creative process.

Understanding how generative AI influences creative ideation in educational contexts is essential for shaping future pedagogical practices. As AI tools become increasingly embedded in academic and professional environments, it is crucial to determine not only their potential to support creativity but also the risks they may pose to originality, autonomy, and diversity of thought. By providing empirical evidence on how timing, interaction, and AI-student dynamics affect creative outcomes, this study contributes to the responsible integration of AI in higher education. The findings aim to inform educators, designers, and policymakers seeking to cultivate more intentional, equitable, and creativity-enhancing uses of AI.

1.1. Theoretical framework

1.1.1. Creativity and education

Creativity, defined as the ability to generate novel and useful ideas (Amabile, 1983; Runco & Jaeger, 2012), has driven human evolution and scientific progress throughout history (Mumford et al., 2012). Since today's students will be responsible for advancing future scientific, technological, and social progress, education systems must ensure the development of competencies related to innovation, adaptability, and creative thinking (Henriksen et al., 2018; United Nations, 2023). Among these, creativity is highlighted as one of the essential 21st-century skills (Cropley, 2020; Newton & Newton, 2014), particularly within engineering and design disciplines, where it supports the creation of sustainable, efficient, and advanced solutions (Badran, 2007; C.-K. Chen & Hsu, 2006; Cropley, 2020). Therefore, these careers must equip students to address complex, open-ended problems by fostering original and divergent thinking (Cropley, 2020).

In academic research, creativity has been extensively examined across various disciplines, including psychology, education, neuroscience, and innovation management, to understand its development, assessment, and applications. Foundational scholars such as Amabile (1983); Guilford (1950), and Torrance (1972) have shaped the theoretical basis for divergent thinking and the development of creative potential. Their work emphasises that creativity is not a fixed trait but a dynamic and multifaceted capacity that can be cultivated. More recent perspectives have expanded this understanding by introducing the concept of distributed creativity, which holds that creativity emerges from interactions among individuals, tools, environments, and cultural contexts (Sawyer & DeZutter, 2009). This view aligns with theories of embodied cognition, which suggest that creative thought is grounded in bodily experience and sensorimotor engagement, and that interactions with physical spaces, objects, and other people contribute to the generation of creative ideas (Foglia & Wilson, 2013; Stanciu, 2015). Furthermore, sociocultural approaches emphasise that creative expression is shaped by shared practices, cultural traditions, social norms, and collaborative dynamics, positioning creativity not only as an individual attribute but as a socially and contextually mediated phenomenon (Glăveanu, 2010).

1.1.2. Collaboration, co-creativity and AI

Grounded in theories that define creativity as socially situated and context-dependent rather than as a fixed personal trait, research has increasingly examined how creative potential can be cultivated through interaction and supportive environments that promote the development and exchange of ideas.

Among these strategies, collaboration has received particular support as a means of enhancing creativity through shared processes of ideation, discussion, and feedback. Collaborative settings can help generate novel perspectives, challenge assumptions, and expand the range of possible solutions. Studies show that working in pairs or groups can improve creative outcomes compared with working individually (Choudhary et al., 2022; Wu & Chen, 2022; Zhou et al., 2020), though others warn that exposure to others' ideas may induce fixation and limit exploration (Kohn & Smith, 2011).

Within this context, generative artificial intelligence offers a new form of creative collaboration, where AI is viewed not only as a

tool but as an active agent in the ideation process (Vinchon et al., 2023). Large language models (LLMs), trained on extensive textual data, can produce coherent responses and engage users in open-ended dialogue, providing an intuitive interface for real-time idea generation and refinement (B. Chen et al., 2023; Kasneci et al., 2023). However, since these systems operate through probabilistic predictions rather than genuine understanding, concerns remain regarding the authenticity, novelty, and depth of the content they produce (Kim & Maher, 2023; Koivisto & Grassini, 2023; Moruzzi, 2025).

A central question is whether such systems can support or potentially hinder creativity. While creativity is deeply tied to human emotion, intuition, and social context (Cropley, 2020; Erden, 2010; Higuera Martínez et al., 2021), AI systems may lack embodied awareness, intentionality, and expressive purpose, potentially leading to superficial or culturally and emotionally detached outputs (Kim & Maher, 2023; Koivisto & Grassini, 2023; Moruzzi, 2025). Some scholars argue that AI cannot be creative in the human sense and caution that overdependence on such systems may inhibit spontaneous creative moments rooted in human intuition and interaction (H. K. Lee, 2022; Wingström et al., 2022).

In light of these concerns, the concept of co-creativity has been reframed, positioning AI as a collaborative partner that complements, rather than replaces, human contributions through hybrid partnerships (Moruzzi, 2025; Vinchon et al., 2023; Wingström et al., 2022). According to Moruzzi (2025), the central question should not be whether AI can be considered creative, but how it can reshape human creative processes and what implications this transformation entails.

1.1.3. Creativity dimensions

Assessing how AI transforms co-creative processes requires a robust framework for evaluating creativity. Classic models such as the Torrance Tests of Creative Thinking (TTCT) (Torrance, 1966), Guilford's Alternative Uses Test (AUT) (Guilford, 1967), and the Wallach-Kogan Creativity Tests (Wallach & Kogan, 1965) have shaped the way creativity is conceptualised and measured. Across these models, four dimensions have become standard in assessing divergent thinking: fluency (ability to generate numerous ideas), flexibility (capacity to shift between different categories or perspectives), originality (production of statistically uncommon or unconventional ideas), and elaboration (degree of refinement and detail in the ideas). These dimensions are widely used in psychological, educational, and design research as reliable indicators of creative performance (Hubert et al., 2024; Koivisto & Grassini, 2023; Moon et al., 2024; Zhao et al., 2025).

Although research on AI's role in creative contexts is still emerging, early studies suggest that it can influence key dimensions of creativity, yet findings are not fully aligned. Some studies report that AI assistance enhances idea fluency, diversity, and detail, sometimes outperforming humans in originality and elaboration (Habib et al., 2024; Hubert et al., 2024; Wieland et al., 2022). However, other studies have found that humans outperform GPT-3 in flexibility and originality, with the most distinctive and unconventional ideas coming from the highest-performing human participants (Koivisto & Grassini, 2023; Stevenson et al., 2022). In addition, Rick et al. (2023) reported that some participants expressed concerns about the AI's tendency to stay close to the initial problem framing, limiting exploration of more unconventional or abstract directions. These mixed results point to the need for further research to clarify the conditions under which AI supports or constrains creative performance.

Beyond these four established dimensions, novelty is often identified as an essential element of creativity. Runco and Jaeger (2012) argued that an idea cannot be considered truly creative unless it is novel or unique. In the context of AI-generated outputs, novelty is particularly relevant, as LLMs rely on existing data patterns, which may constrain their ability to produce genuinely innovative ideas. However, the distinction between novelty and originality is often blurred, with the terms frequently used interchangeably. Despite the lack of a standard metric and its rare treatment as a separate dimension, some studies have attempted to assess it in AI-assisted contexts. For example, Chandrasekera et al. (2024) used a scale that included novelty to evaluate concepts generated with and without a text-to-image AI tool, though separate scores were not reported. Boussioux et al. (2024) found that human participants generated ideas with greater novelty than ChatGPT, with novelty understood as the degree of divergence from existing solutions. To address this ambiguity, the present study treats novelty as an independent dimension, allowing for a more refined evaluation of whether AI-assisted ideation contributes genuinely new ideas.

1.1.4. Homogenisation, timing, and human-ai interaction

Beyond traditional assessments, emerging research on AI co-creativity highlights the need to examine underexplored factors that may shape creative outcomes.

One such factor is the timing of AI integration in the ideation process. Some studies show that initial AI support can leave a residual benefit, with users maintaining self-regulated behaviours even after assistance is withdrawn (Chandrasekera et al., 2024; Wang et al., 2025). This aligns with theories of scaffolding and fading (Vygotsky, 1978; Wood et al., 1976), which hold that temporary support enables learners to perform beyond their current abilities before being gradually withdrawn to foster autonomy. Still, others caution that early AI input may impede creative autonomy, leading to fixation and reduced ability to generate ideas unaided (Habib et al., 2024). These contrasting findings suggest that the impact of early AI use on subsequent ideation tasks remains inconclusive. This study builds on previous findings by examining whether initial AI support enhances later performance and how the timing of AI integration interacts with task conditions to affect creativity.

Another emerging concern is the homogenising effect of LLMs on idea generation. Since LLMs draw on statistical patterns from training data, their outputs may converge around similar themes and expressions (Baltà-Salvador et al., 2025; Doshi & Hauser, 2024; Ivcevic & Grandinetti, 2024; Zhao et al., 2024). Empirical analyses using semantic distance measures have shown that AI-assisted ideas and texts tend to exhibit greater uniformity than those generated by humans, even when creative prompts or parameter adjustments, such as frequency penalties, are applied (Anderson et al., 2024; Moon et al., 2024). Therefore, while such tools can enhance individual performance, their widespread use may standardise the outputs, reducing collective creative diversity. The present study extends

previous research on the homogenising effects of generative AI by employing additional analytical methods to provide a more comprehensive assessment of its impact on output diversity.

Finally, another relevant line of inquiry concerns how users perceive and engage with AI tools during creative tasks. Research suggests that students often treat AI as a functional assistant rather than a creative or relational partner, leading to largely passive engagement (Marrone et al., 2022). Studies report that this perception, combined with an increasing reliance on AI, can reduce decision-making and critical thinking, thereby diminishing the role of human cognition in academic contexts and fostering cognitive laziness (Ahmad et al., 2023; Kasneci et al., 2023). Building on these insights, this study examines student-AI interactions through a qualitative analysis of prompts, aiming to understand the relational and conversational dynamics that unfold during co-creative practices.

2. Present study

This study examines co-creativity between students and AI in the context of a ChatGPT-assisted brainstorming task. The research focuses on understanding how generative AI tools can influence the ideation process and shape creative outcomes. The study's research questions are listed below, each accompanied by text that clarifies the specific focus of the investigation, outlines the rationale and anticipated hypotheses, and links the question to the relevant literature:

1. **Does the use of AI influence the creativity of ideas produced by students?** This question examines whether students generate more creative ideas with AI assistance compared to working unaided. Creativity was evaluated along five dimensions: fluency, flexibility, originality, elaboration, and novelty. In line with previous findings suggesting AI enhances creative performance (Habib et al., 2024; Hubert et al., 2024; Li et al., 2024; Wieland et al., 2022), it is hypothesised that AI-supported ideation would outperform the unaided condition across all dimensions.
2. **Does the order in which AI is used during tasks affect creative outcomes?** This question explores the interaction between the condition (AI-assisted vs. unaided) and the order of exposure (AI-first vs. non-AI-first). It aims to investigate whether the timing of AI integration impacts students' creative performance across tasks. Drawing on research into fading and residual effects of early AI use in learning processes (Chandrasekera et al., 2024; Wang et al., 2025), it is hypothesised that students who begin with AI support would perform better on subsequent unaided tasks.
3. **Does the use of AI result in a homogenisation of the ideas generated?** To address this question, two forms of homogenisation were considered: thematic homogenisation, referring to convergence around similar topics or categories, and semantic homogenisation, referring to convergence in textual expression across ideas.
   - Thematic homogenisation was assessed by examining whether AI-generated ideas were distributed across fewer categories, clustered within dominant themes, or exhibited a lower frequency of unique contributions.
   - Semantic homogenisation was evaluated at both group and individual levels by comparing the semantic similarity of idea descriptions in the AI-assisted and unaided conditions.

   While numerous studies emphasise the benefits of AI in enhancing creative ideation (Habib et al., 2024; Hubert et al., 2024; Li et al., 2024; Wieland et al., 2022), others caution that its use may lead to homogenisation of ideas (Anderson et al., 2024; Doshi & Hauser, 2024; Ivcevic & Grandinetti, 2024; Moon et al., 2024; Zhao et al., 2025). Building on these concerns, this study provides new insights into whether AI-assisted creativity contributes to idea convergence.
4. **How do students interact with AI tools during the ideation process?** An exploratory qualitative analysis was conducted on the **conversations** generated during AI-assisted tasks. The goal was to identify interaction patterns, examining whether students used AI functionally or relationally and whether signs of dependence, critical engagement, or collaborative exploration emerged. This sheds light on the spontaneous integration of AI into creative workflows and its implications for student learning and engagement.

3. Method

3.1. Participants

The study involved 51 third-year students from the bachelor's degree in industrial design engineering and product development at the Universitat Politècnica de Catalunya · BarcelonaTech (UPC), Spain. As third-year students, participants had the necessary prior knowledge and experience in idea generation and creative processes to perform the tasks effectively.

3.2. Procedure

3.2.1. Controlled experiment

To conduct the research, a randomised controlled crossover experiment was employed, in which participants were randomly assigned to one condition for the first task and then switched to the other condition for the second task. This design allowed each participant to serve as their own control, reducing inter-individual variability and enhancing the statistical power of paired comparisons between AI-assisted and unaided ideation. Moreover, alternating the starting condition enabled an examination of whether the order in which AI was introduced influenced subsequent performance.

The experiment took place during a two-hour session in which participants were asked to brainstorm innovative ideas. All students first gathered in a lecture hall, where the procedure was explained. They were then randomly divided into two groups and directed to

separate classrooms:

- **Unaided**: Participants in this condition were placed in a classroom without computer access or permission to use any external resources or electronic devices.
- **AI-assisted:** Participants in this condition worked in a computer lab using university-owned computers with standardised equipment and limited internet access to ChatGPT (GPT-3.5).

In their assigned rooms, participants completed the first task (T1), which involved generating innovative ideas for improving public transport. They had 30 min to produce eight ideas, each documented on a separate sheet using pre-printed templates with space for drawings and descriptions. Each sheet included a unique ID to ensure anonymity during analysis. Throughout the task, participants were supervised to ensure they followed the instructions.

After 30 min, supervisors collected the completed sheets, and participants switched conditions and classrooms for the second task (T2), as illustrated in Fig. 1. In T2, participants generated ideas for improving university spaces, following the same procedure as in T1. The two tasks were designed to be comparable in terms of difficulty, scope, and relevance.

In total, 728 ideas were gathered. This provided a large enough dataset to conduct reliable analyses of emerging patterns, trends, and significant differences.

3.2.2. Idea categorisation

To support the analysis of thematic homogenisation and the assessment of some creativity dimensions, a categorisation process was applied to the 728 ideas generated during the experiment. To ensure impartiality, all ideas were completely anonymised, preventing those involved in categorisation from accessing participant identities or knowing whether AI had been used.

Using a Qualitative Content Analysis (QCA) approach, a multilevel classification system was developed inductively based on the dataset. The process began with a data-driven open coding, in which each idea was treated as a distinct unit and assigned a descriptive label based on its core topic, problem, or proposed solution. To ensure analytical consistency, codes were mutually exclusive to avoid duplication. In cases where an idea could fit into multiple categories, it was agreed to assign it to the category most representative of its primary concept. This approach ensured clarity, prevented analytical fragmentation and overlap, and maintained consistency across the coding process. The resulting codes were reviewed for similarities, contrasts, and conceptual relationships.

These initial codes were then iteratively grouped into broader categories through comparison and refinement, resulting in a three-level hierarchical structure that enabled analysis at varying levels of granularity. All category names and structures emerged from patterns within the data, rather than imposed externally. Throughout the process, categories were refined, split, or renamed to ensure internal consistency, better reflect conceptual distinctions, and maintain overall coherence. The categorisation was peer-reviewed to assess coherence and completeness. Discrepancies were resolved collaboratively through discussion among the researchers until consensus was reached, with final adjustments made to ensure that the system accurately represented the dataset. This emphasis on discussion-based resolution, rather than on quantifying agreement, reflects Barbour's (2001) observation that the content of disagreements and the insights generated by such discussions are valuable for refining the coding framework.

The final version of the hierarchical categorisation reflected three increasing levels of specificity:

- **First-Level (LV1):** Broad thematic domains identified for each task (e.g., in Task 2: *classrooms, common areas, services*, and *campus*).
- **Second-Level (LV2)**: More specific categories nested within each LV1 theme (e.g., under classrooms: *furniture comfort, technology, layout*, and *storage*).
- **Third-Level (LV3)**: Subcategories reflecting the same needs or problems addressed within each LV2 category (e.g., within layout: *modular furniture, circular classrooms, dividers, and foldable tables*).

To assess originality, two additional classifications reflecting unique contributions were defined:

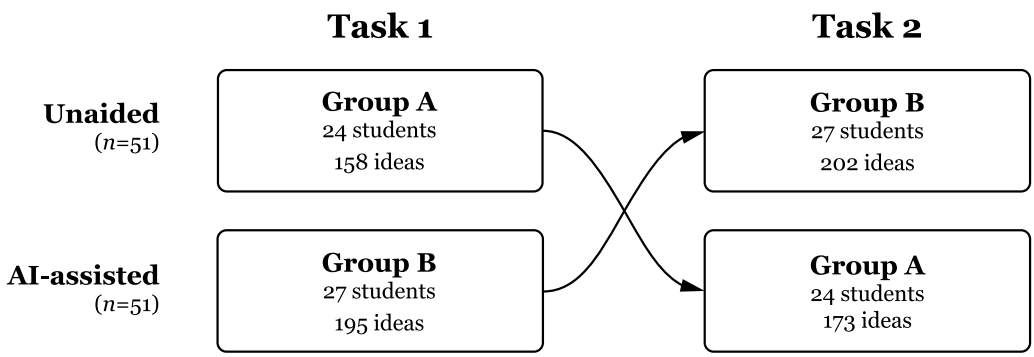

Fig. 1. Overview of the experiment, participants, and outcomes.

- **Unique Ideas (UI):** Ideas that did not align with any existing LV3 subcategory, meaning no other idea addressed the same specific problem or need.
- **Unique Solutions (US)**: Within each LV3 subcategory, ideas that addressed the same problem but proposed clearly different **solutions** were considered unique solutions.

To assess novelty, ideas resembling existing products were labelled as Existing Solutions (ES).
Table 1 summarises the categories, unique contributions and existing solutions identified across tasks.

3.2.3. Creativity assessment instrument

To evaluate the impact of AI on ideation outcomes, creativity was assessed across five key dimensions: fluency, flexibility, originality, elaboration, and the study-specific addition of novelty. These metrics were adapted from established creativity assessments (Guilford, 1967; Runco & Jaeger, 2012; Torrance, 1966; Wallach & Kogan, 1965). Since this study focused on the creativity of the outputs rather than the individual creative abilities of the participants, the measures were adapted to suit the specific context of this research. A small-scale pilot test was conducted to assess the clarity and applicability of the metrics.

As outlined in the theoretical framework, the study extends beyond evaluating originality based on frequency by also examining whether ideas represented genuinely novel contributions, distinct from existing products or known solutions. To capture this, the evaluation instrument included novelty as a separate dimension. Although the concept of novelty is well-established in the literature (Runco & Jaeger, 2012), methods for measuring it vary across domains, with few standardised approaches and frequent conceptual overlap with originality. For this reason, a tailored measure was developed specifically for this study, grounded in theoretical definitions and aligned with its context and objectives.

Table 2 presents the final instrument developed to assess the creativity of the outcomes, detailing the metrics and measurement methods for each.

3.3. Analysis

A mixed-methods approach was employed, combining quantitative assessments of creativity and diversity with qualitative analysis of student-AI interactions. First, normality was tested via the Shapiro-Wilk test. Since all variables except image elaboration deviated from normality ($p < .05$), non-parametric tests were applied. Effect sizes were reported using Cohen's (1988) criteria.

3.3.1. Analysis of AI impact on creativity dimensions and order effects

To compare creative performance across AI-assisted and unaided conditions, a Wilcoxon signed-rank test was conducted on the creativity dimensions. As a preliminary step to ensure group comparability, a Mann-Whitney U test was performed to confirm that no significant differences existed. This analysis was based on students' performance in five prior assignments (A1-A5), all of which involved creativity and ideation skills, as well as their cumulative GPA up to that point in their bachelor's studies.

A comparison was conducted to assess differences between students who began the task with AI support (Group B) versus those who started without it (Group A). First, the Mann-Whitney U test was used to compare creativity scores across tasks. Then, a mixed between-within subjects ANOVA was conducted with ideation condition (with vs. without AI) as the within-subjects factor and group order (AI-first vs. non-AI-first) as the between-subjects factor.

Although ANOVA assumes normality, the test is generally robust to mild deviations from this assumption (Pallant, 2020). To further assess the assumptions, several diagnostics were conducted. For all variables except fluency, Shapiro-Wilk W values were high (many above 0.95), indicating only slight deviations from normality. Furthermore, skewness and kurtosis values were below 1.0, suggesting acceptable symmetry and distribution shape. Levene's test and Box's M test both yielded non-significant results ($p > .05$), supporting the assumptions of homogeneity of variances and equality of covariance matrices, respectively. Given the reasonably sized and balanced groups ($n > 20$ per group) and the generally acceptable distribution characteristics, it was concluded that the assumptions for conducting ANOVA were sufficiently met for these variables. Fluency was the only variable that clearly deviated from these assumptions, with a Shapiro-Wilk W value below 0.70, skewness above 2.0, and kurtosis near 5, indicating substantial asymmetry and heavy tails. Levene's test ($p < .05$) and Box's M test ($p = .001$) also indicated violations of the homogeneity and covariance assumptions. Therefore, results involving fluency should be interpreted with caution.

3.3.2. Evaluation of thematic and semantic homogenisation

The homogenisation of the results was examined using two sets of complementary analyses to capture what students conceptualised and how they expressed those ideas.

Table 1
Categories, unique ideas, unique solutions, and existing solutions by task.

	LV1	LV2	LV3	UI	US	ES
Task 1	4	29	68	90	60	45
Task 2	4	26	75	76	91	34

Note. LV1 = First-Level category; LV2 = Second-Level categories; LV3 = Third-Level subcategories; UI = Unique Ideas; US = Unique Solutions; ES = Existing Solutions.

Table 2
Creativity instrument metrics and measurements.

Metric	Description	Measurement
Fluency	Number of ideas generated	Total number of ideas generated
Flexibility	Diversity of conceptual categories explored	Second-Level categories explored
Originality	Rarity of ideas relative to the full dataset	Count of Unique Ideas and Unique Solutions
Novelty	The extent to which ideas are novel, disruptive, or go beyond existing solutions	Count of Existing Solutions. A lower number reflects greater novelty
Text Elaboration	The detail and clarity of written idea descriptions	Word count of the idea description
Image Elaboration	The visual detail and representational richness	Representation of human interaction (Yes 1 / No 0) Representation of motion or states (Yes 1 / No 0) Representation of 3D volume (Yes 1 / No 0)

Thematic homogenisation was assessed by analysing how ideas were distributed across categories. Analyses included:

- Visual inspection of the distribution of AI-assisted and unaided ideas across categories and descriptive statistics of the categories and unique contributions.
- Calculation of Shannon and Simpson diversity indices to evaluate both the richness and evenness of categories. Results followed the interpretation of Baliton et al. (2020) and Guajardo (2015).
- Comparison of average category sizes of AI-assisted and unaided ideas to assess whether AI-generated ideas clustered in larger categories, using a Mann-Whitney U test.
- Examination of whether AI use was associated with differences in the number of Unique Ideas and Unique Solutions, using chi-square tests.

Semantic homogenisation was evaluated through a semantic divergence analysis of the idea descriptions. Text data from student submissions were extracted using OCR through Gemini AI, followed by manual verification. Each idea was encoded using LaBSE (Language-agnostic BERT Sentence Embedding), a multilingual transformer-based model.

- At the group level, the cosine distance between each idea and the average embedding of its respective group (AI-assisted vs. unaided) was calculated and compared using an independent samples *t*-test.
- At the individual level, the average pairwise semantic divergence within each participant's set of ideas was computed per condition (AI-assisted vs. unaided) and compared via a paired samples *t*-test.

3.3.3. Qualitative analysis of student-ai interactions

To explore patterns in student-AI interaction, a descriptive analysis was first conducted, examining prompt frequency, word count, and code diversity.

This was followed by a qualitative thematic analysis of the prompts generated during AI-assisted tasks, using a bottom-up, latent approach to identify underlying communicative patterns beyond the surface content. The process began with a close reading of the conversations to familiarise with recurring behaviours and intents. Prompts containing multiple sentences or distinct communicative intentions were segmented into separate units to allow for more precise coding and avoid overlap. Initial codes were developed inductively to capture specific interactional purposes, with each unit assigned a primary intent code. The coding scheme was iteratively refined through peer debriefing to clarify definitions, adjust thematic boundaries, and ensure consistency across the dataset. Ambiguities were resolved through consensus, achieved via discussion and re-examination of the data.

Verbatim quotes are presented in the Results section to enhance transparency and understanding. Additionally, interpretive reflections that emerged during the analysis are included in the Discussion section to provide deeper insights beyond the coding of the conversations.

Table 3
Baseline comparison of academic performance between groups.

	A1	A2	A3	A4	A5	GPA
Group A *M(SD)*	7.58 (1.34)	7.84 (1.20)	8.37 (0.92)	8.08 (1.08)	7.93 (1.23)	7.15 (0.58)
Group B *M(SD)*	7.56 (1.87)	7.64 (1.37)	8.53 (0.88)	8.42 (1.06)	8.19 (1.51)	7.19 (0.55)
Mann-Whitney U test						
U	350	294	353	385	364	326
z	0.49	−0.58	0.55	1.16	0.76	0.04
p	.623	.565	.583	.246	.447	.970

Note. $N = 51$; A1-A5 = Assignments 1–5; GPA = Grade Point Average; Group A $n = 24$; Group B $n = 27$; M = mean; SD = standard deviation; U = Mann-Whitney U test statistic; z = standardised test statistic; p = significance level.

4. Results

4.1. Wilcoxon signed-rank test results on creativity dimensions

The results of the Mann-Whitney U test indicated no significant differences between the two groups in prior academic performance, including five assignments (A1-A5) and cumulative GPA, suggesting that the random assignment did not introduce bias (Table 3).

With baseline equivalence established, a paired-sample test was conducted to examine whether AI assistance had a significant impact on students' creativity outcomes. As shown in Table 4, the Wilcoxon signed-rank test revealed significant differences in the dimensions of novelty and text elaboration. Specifically, students produced less detailed descriptions and generated more ideas that resembled existing solutions when working with AI support. According to Cohen's (1988) guidelines, the effect sizes for these differences were moderate ($0.30 < r < 0.50$), suggesting that AI had a meaningful impact on these aspects of creativity. No significant differences were found for the remaining creativity metrics.

4.2. Mann-Whitney U test and mixed Anova results on task order effects

To examine performance across tasks and creativity dimensions, a Mann-Whitney U test was conducted to compare both groups. As shown in Table 5, Group B, who used AI during T1 and completed T2 without it, scored significantly higher in both flexibility and originality across both tasks compared to Group A, who completed T1 without AI and used it for T2. Notably, Group B outperformed Group A in T2 despite not using AI, suggesting a possible residual learning effect in which prior exposure to the AI tool influenced later performance. Effect sizes ranged from $r = 0.31$ to 0.40, indicating moderate to moderately large effects.

To further explore the interaction between AI use and condition order on creativity outcomes, a mixed between-within subjects ANOVA was conducted. Results revealed a significant interaction between AI usage and condition order for flexibility (*Wilks'* $\lambda = 0.801$, $F(1, 49) = 12.19$, $p = .001$, $\eta^2_p = 0.20$) and originality (*Wilks'* $\lambda = 0.747$, $F(1, 49) = 16.62$, $p < .001$, $\eta^2_p = 0.25$). Therefore, the order in which participants encountered the AI condition significantly influenced their performance on these dimensions. Based on Cohen's (1988) criteria, both effects were large ($\eta^2_p > 0.14$), accounting for a substantial portion of the variance in flexibility and originality scores. No significant interaction effects were found for the remaining dimensions.

Regarding novelty, a significant main effect of AI usage was found (*Wilks'* $\lambda = 0.872$, $F(1, 49) = 7.19$, $p = .010$, $\eta^2_p = 0.13$), representing a moderate-to-large effect size. In contrast, there was no significant main effect of condition order ($F(1, 49) = 1.56$, $p = .218$, $\eta^2_p = 0.03$). These findings suggest that the use of AI led to a greater number of ideas based on existing solutions, regardless of when it was introduced.

4.3. Diversity indices, mann-whitney U, chi-square, and semantic divergence results on homogenisation effects

Thematic homogenisation was first explored through a visualisation of the categorisation results to identify patterns in idea distribution. The hierarchical structure resulting from idea categorisation is shown in Fig. 2, where the Y-axis displays the First-Level (LV1) themes and the X-axis shows the total number of ideas per theme. Within each bar, segments represent Second-Level (LV2) categories, with their size corresponding to the number of ideas they contain. Segment colours indicate the proportion of ideas generated with AI assistance, allowing for visual comparison across LV2 categories.

As can be observed, categories with the highest proportion of AI-assisted ideas were not the largest ones. In fact, the largest categories contained a mix of both AI-assisted and unaided ideas, while the smallest categories tended to be composed entirely of either AI-assisted or unaided ideas. Notably, some LV2 categories emerged exclusively from AI-assisted ideas, suggesting that AI contributed to the exploration of areas that might not have been considered otherwise. Conversely, other categories were composed only of unaided ideas, indicating areas where human ideation remained distinct.

Table 6 complements the previous figure by showing the number of LV2 and LV3 categories containing only AI-assisted ideas, only unaided ideas, or a mix of both, along with the total number of Unique Ideas, Unique Solutions and Existing Solutions derived from the 728 ideas collected.

Table 4
Comparison of creativity dimensions in Ai-assisted and unaided conditions.

	Fluency	Flexibility	Originality	Novelty	Text Elaboration	Image Elaboration
Unaided *M(SD)*	7.06 (1.54)	5.51 (1.78)	3.27 (1.74)	0.55 (0.81)	25.57 (8.06)	1.32 (0.36)
AI-assisted *M(SD)*	7.22 (1.08)	5.57 (1.36)	2.94 (1.35)	1.00 (0.96)	23.36 (7.66)	1.28 (0.40)
Wilcoxon signed-rank test						
z	0.77	0.21	−1.24	2.55	−2.44	−0.18
p	.439	.833	.216	.011*	.015*	.857
r	.11	.03	.17	.36	.34	.03

Note. $N = 102$; Unaided $n = 51$. AI-assisted $n = 51$. M = mean; SD = standard deviation; z = standardised test statistic; p = significance level; r = effect size.
* $p < .05$. ** $p < .01$. *** $p < .001$.

Table 5
Comparison of creativity dimensions in AI-assisted and unaided conditions across tasks.

		Fluency	Flexibility	Originality	Novelty	Text Elaboration	Image Elaboration
Task 1							
Group A *M(SD)*	Unaided	6.58 (1.98)	4.58 (1.81)	2.50 (1.47)	0.67 (0.87)	25.54 (8.92)	1.35 (0.35)
Group B *M(SD)*	AI-assisted	7.22 (1.12)	5.67 (1.54)	3.33 (1.07)	1.07 (1.07)	25.19 (7.14)	1.20 (0.43)
Mann-Whitney U test							
U		374	440	437	392	323	260
z		1.03	2.21	2.20	1.36	0.03	1.22
p		.304	.027*	.028*	.172	.977	.223
r		.14	.31	.31	.19	.00	.17
Task 2							
Group A *M(SD)*	AI-assisted	7.21 (1.06)	5.46 (1.14)	2.50 (1.50)	0.92 (0.83)	21.31 (7.85)	1.36 (0.36)
Group B *M(SD)*	Unaided	7.48 (0.85)	6.33 (1.30)	3.96 (1.70)	0.44 (0.75)	25.60 (7.39)	1.29 (0.36)
Mann-Whitney U test							
U		373	461	474	211	444	279
z		1.06	2.65	2.88	−2.35	2.26	0.86
p		.289	.008**	.004**	.019*	.024*	.389
r		.15	.37	.40	.33	.32	.12

Note. $N = 51$; Group A $n = 24$. Group B $n = 27$. M = mean; SD = standard deviation; U = Mann-Whitney U test statistic; z = standardised test statistic; p = significance level; r = effect size.
* $p < .05$. ** $p < .01$. *** $p < .001$.

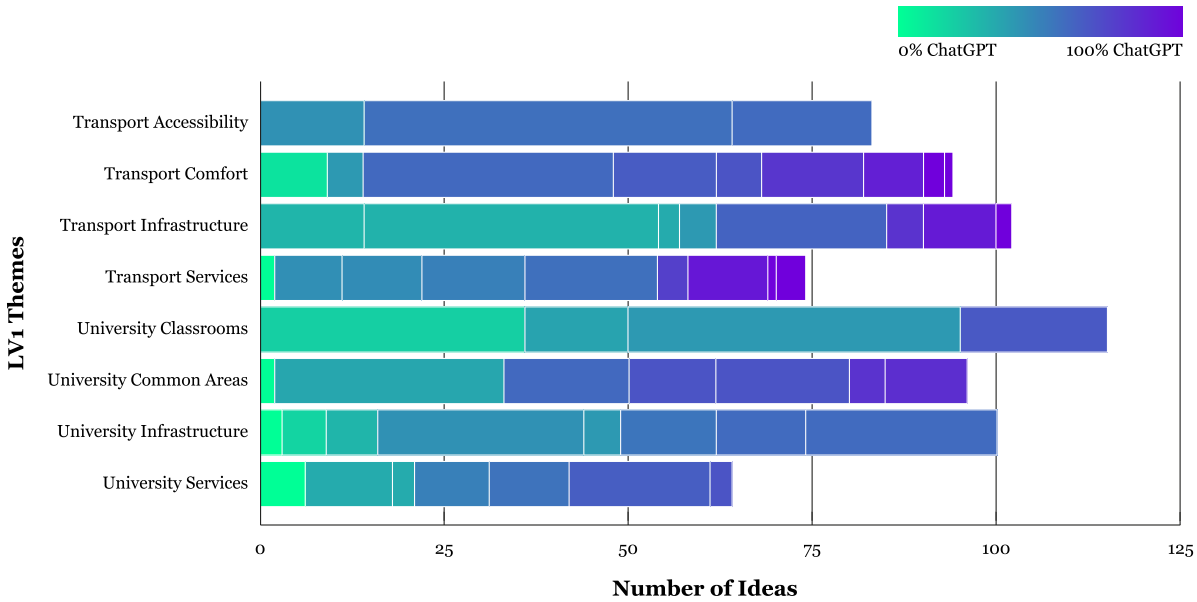

Fig. 2. Distribution of ideas within LV1 themes and LV2 categories.

Table 6
Distribution of idea categories and unique contributions by condition.

	LV2	LV3	UI	US	ES
Unaided	4	26	85	82	28
Mixed	46	88	-	-	-
AI-assisted	5	29	81	69	51

Note. LV2 = Second-Level categories; LV3 = Third-Level subcategories; UI = Unique Ideas; US = Unique Solutions; ES = Existing Solutions. Mixed = Categories that included both AI-assisted and unaided ideas.

A quantitative assessment using the Shannon and Simpson indices evaluated the diversity and distribution of ideas across categories. This aimed to detect measurable patterns of homogenisation between AI-assisted and unaided conditions. As shown in Table 7, ideas from both groups were distributed across a similar number of categories at both levels. Shannon index values above 3.5 for both groups indicated broad category distribution, high evenness, and no dominant category. Similarly, the Simpson index showed very high diversity in both groups, in some cases nearing complete heterogeneity. These results suggest that ideas generated with and without AI assistance displayed comparable thematic diversity, even with the AI-assisted ideas showing a slightly broader and more even distribution.

To further examine homogenisation, an analysis was conducted to determine whether AI-assisted ideas were more likely to cluster in larger categories and whether they differed in the number of unique contributions produced. Category size was calculated to assess whether AI-assisted ideas (N= 368, M= 4.22, SD = 3.01) were more likely to appear in larger categories compared to unaided ideas (N= 360, M= 4.58, SD = 3.65). The results showed no significant difference in category size (U= 65,489, z = −0.27, p = .798), suggesting AI-assisted ideas did not converge more heavily in dominant thematic clusters.

Moreover, no association was found between the use of AI and the total number of unique contributions ($\chi^2(1, N= 728) = 2.12, p = .145$). A more detailed analysis also revealed no significant differences between conditions in the number of Unique Ideas ($\chi^2(1, N= 728) = 0.18, p = .670$) and Unique Solutions ($\chi^2(1, N= 728) = 1.56, p = .212$) when examined separately, suggesting that AI use did not reduce the number of unique contributions.

Semantic homogenisation was examined through the analysis of semantic divergence at the group and individual levels. Fig. 3 presents the distribution of semantic divergence for both conditions and levels of analysis.

At group level, the average semantic divergence to the centroid was M= 0.28 (SD = 0.06) for unaided ideas and M= 0.29 (SD = 0.07) for AI-assisted ideas. An independent samples t-test revealed a statistically significant difference between the two conditions, $t(728) = -3.54, p < .001, d = -0.26$, 95 % CI [−0.03, −0.01]. According to Cohen's (1988) guidelines, the effect size was small ($0.20 < d < 0.50$) suggesting that AI-assisted ideas were modestly more semantically similar to the group centroid, which points to a potential homogenising effect at the group level.

At individual level, the average semantic divergence from the centroid per participant was M= 0.28 (SD = 0.03) for unaided ideas and M= 0.29 (SD = 0.03) for AI-assisted ideas. A paired samples t-test showed a statistically significant difference, $t(364) = 3.41, p = .001, d = 0.48$, 95 % CI [0.01, 0.03]. In this case, the effect size approaches the medium range, indicating that participants generated more semantically concentrated ideas when using AI, which further supports the presence of a homogenising effect at the individual level.

4.4. Thematic coding and descriptive analysis of student-AI interactions

To address the last research question, students' conversations with ChatGPT were coded. Of the 51 participants, 49 conversations were successfully retrieved, yielding 341 units of analysis. A statistical analysis was conducted to examine the number of prompts per conversation, the word count per prompt, and the number of codes per conversation (Table 8).

Table 9 summarises the types of interactions students had with ChatGPT during the ideation process, including the frequency of each code and illustrative examples of the prompts.

5. Discussion

This study offers valuable insights into the role of generative AI in ideation, revealing patterns of human-AI collaboration that challenge overly simplified views of its benefits and limitations. The findings underscore the need for educational institutions to adopt purposeful strategies that replace passive AI integration with creative, critical, and reflective engagement.

5.1. Creativity outcomes in AI-Assisted vs. unaided conditions

Contrary to expectations, the use of AI did not significantly affect fluency, flexibility, or originality. Students generated a comparable number of ideas across conditions, with similar category diversity and frequency of unique contributions. These findings challenge claims that AI suppresses original thought and creativity (Koivisto & Grassini, 2023; Stevenson et al., 2022). At the same time, they also question whether AI enhances creative outcomes, as performance remained consistent between AI-assisted and unaided tasks (Habib et al., 2024; Hubert et al., 2024; Li et al., 2024; Wieland et al., 2022).

However, differences emerged in novelty and elaboration dimensions. While the quantity and diversity of ideas were similar across

Table 7
Diversity indices of idea categories by AI condition.

	LV2	Shannon Index (*H'*)	Simpson Index (*1-D*)	LV3	Shannon Index (*H'*)	Simpson Index (*1-D*)
Unaided	50	3.538	0.964	114	4.503	0.990
AI-assisted	51	3.661	0.972	117	4.588	0.992
Difference		−0.123	−0.008		−0.085	−0.002

Note. LV2 = Second-Level categories; LV3 = Third-Level subcategories.

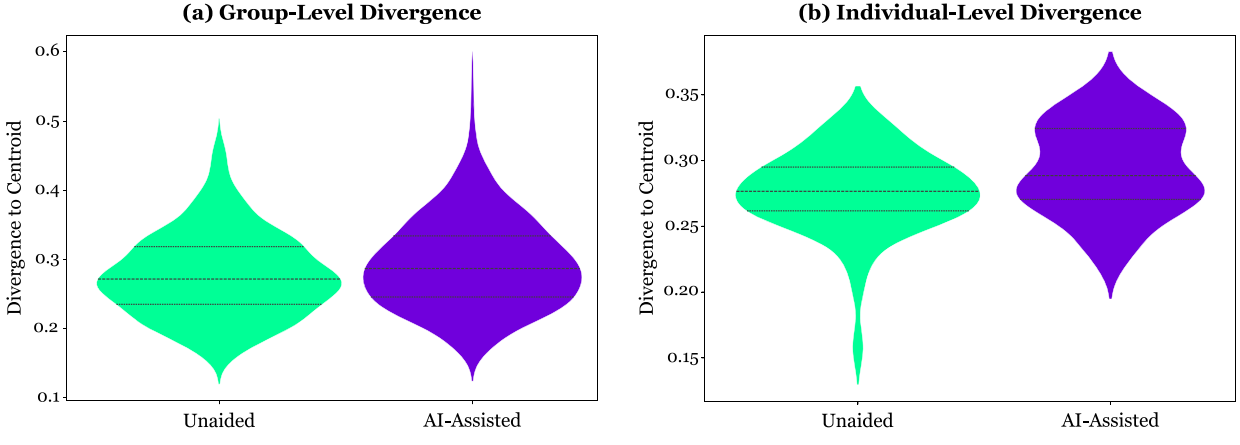

Fig. 3. (a) Group-level divergence in idea similarity. (b) Individual-level divergence in idea similarity.

Table 8
Descriptive statistics of student prompts during AI-assisted ideation.

Metric	Prompts per Conversation	Words per Prompt	Codes per Conversation
N	341 prompts	341 prompts	49 conversations
Mean	6.96	9.46	2.31
Standard Deviation	4.18	8.00	1.02
Minimum	1	1	1
25th Percentile	4	5	2
Median (50th Percentile)	6	8	2
75th Percentile	9	11	3
Maximum	22	84	6
Mode	4	8	2

Table 9
Student-AI interaction codes and frequencies during AI-assisted ideation.

Code	*n* (%)	Description	Quote
Generic Idea Generation	**165 (48.39)**	Request for multiple innovative ideas related to the task, without specifying the content or objectives	*"Give me innovative ideas to improve university spaces"* *"I need to brainstorm innovative ideas to improve public transport"*
Specific Ideas	**81 (23.75)**	Specify the content or objective of the idea	*"How would you make a parking lot more efficient for a university?"* *"Give me examples of technological products that could help make public transport more sustainable"*
Useful Information	**40 (11.73)**	Ask questions to gather information related to the topic to better understand the problems or needs	*"What are the main issues in university spaces?"* *"What are the main accessibility challenges in public transport?"*
Idea Development	**29 (8.50)**	Request further details, examples, or clarifications to develop ideas	*"Develop the first idea for me, what would you design?"* *"Explain how the design of a foldable ergonomic chair for outdoor university spaces would be, including functional details"*
Innovation	**11 (3.23)**	Focus on the level of innovation in the ideas	*"Give me crazier ideas"* *"Give me 8 futuristic and creative ideas for public transport that could seem like science fiction"*
Assistance to Write or Draw	**6 (1.76)**	Seek help with how to write or draw the idea effectively	*"For each idea, generate a brief description and how you would briefly draw it so that it can be understood"* *"How would you make the drawing?"*
Opinion	**2 (0.59)**	Ask for an opinion on the idea	*What do you think about having walls that can be painted and graffitied, and periodically repainted white to start over?* *What do you think about having hallways where each one has a theme, and in some, you put the news…*
Other	**7 (2.05)**	Include greetings, expressions of gratitude, or requests for translation	*"Hello chat good afternoon"* *"Thanks man"*

conditions, AI-assisted ideas tended to resemble existing products, indicating a bias towards familiar concepts. ANOVA results confirmed that this was a consistent effect of AI use, rather than a sequencing artefact. This finding can be attributed to the probabilistic nature of large language models, which generate outputs based on patterns derived from prior data (B. Chen et al., 2023; Kasneci et al., 2023). To counter this tendency and avoid anchoring ideation to existing concepts, educators should warn students about AI's bias toward familiar ideas and design activities that encourage unconventional and disruptive exploration, such as imagining impossible solutions, combining unrelated ideas, or reframing problems from radically different perspectives.

A notable decrease in text elaboration was also observed in the AI-assisted condition. One explanation could be that interacting with AI tools diverted students' attention from refining their ideas on paper. Given the fixed time frame for both conditions, time spent engaging with ChatGPT may have reduced opportunities for writing more detailed responses. This suggests that while AI can support idea generation, it may also fragment students' focus, underscoring the need to balance digital and analogue processes in blended creative workflows. To address this, educators should allocate sufficient time for both AI interaction and independent refinement, ensuring that students can engage meaningfully with the tool before fully developing their final outputs.

5.2. Effects of AI introduction order on creative outcomes

A key contribution of this study is the finding that the order in which AI is introduced can significantly influence creative performance. Students who began with AI assistance and then moved to unaided ideation demonstrated higher levels of flexibility and originality. This advantage was maintained in the second task, when they no longer had access to AI, indicating a lasting effect of early AI exposure on subsequent independent performance.

These results align with scaffolding and fading theories (Vygotsky, 1978; Wood et al., 1976), which suggest that temporary support enables learners to internalise effective strategies that they can later apply independently. Initial exposure to AI may introduce students to a wide range of perspectives and approaches to problem-solving, helping to establish a cognitive foundation that enhances originality and flexibility even in the absence of AI. In this sense, AI serves not only as a tool for idea generation but also as a catalyst for sustained cognitive engagement and the transfer of learning (Chandrasekera et al., 2024; Wang et al., 2025).

The implications of these findings may extend beyond ideation. Similar to how training wheels assist learners in mastering bicycle riding, AI may function as an instructional scaffold that helps students manage complex tasks and gradually build the skills and confidence required for independent performance. In practice, educators can harness AI's scaffolding benefits by introducing it early in complex learning tasks and gradually reducing its use to promote independent performance. This approach can provide students with personalised guidance at the outset, helping them develop strategies and problem-solving skills while fostering long-term competencies.

5.3. Patterns of homogenisation in idea generation

The findings indicate that AI use during ideation did not lead to thematic homogenisation, as AI-assisted ideas were not clustered into fewer or broader categories. In fact, the number of categories and the diversity indices suggest that AI-supported ideation was at least as diverse as ideation without AI assistance. Furthermore, there were no significant differences in the number of Unique Ideas or Solutions between the two conditions. These results challenge the concerns that AI may constrain creative exploration or narrow the diversity of the outcomes (Baltà-Salvador et al., 2025; Doshi & Hauser, 2024; Ivcevic & Grandinetti, 2024).

Notably, 29 subcategories and 150 Unique Ideas and Solutions emerged exclusively in the AI-assisted condition. This suggests that AI has the potential to expand the ideation process by introducing new directions that participants might not have otherwise considered. This is particularly relevant, as a diverse set of ideas and a broader spectrum of potential solutions in the early stages of ideation can lead to higher-quality outcomes and more innovative and effective problem solving (Muñoz Adánez, 2005; Paulus et al., 2011). Therefore, rather than replacing human creativity, AI may function as a collaborator that enhances the generation, refinement, and exploration of ideas, serving as a stimulus for creative thinking.

At the same time, 26 categories and 167 unique contributions emerged exclusively from participants working without AI. This demonstrates that both AI-assisted and unaided ideation contribute meaningfully to the creative process. Based on these findings, it is recommended that educators design ideation activities that deliberately combine AI support with autonomous thinking. Such integration can enhance innovation and broaden the exploration of potential ideas, leveraging AI's generative capacity alongside the depth and intuition of human cognition.

While thematic diversity was similar for both conditions, the results diverged concerning semantic homogenisation. At both group and individual levels, semantic divergence was significantly lower in the AI condition. This suggests that although participants explored a wide range of themes, they tended to express their ideas in more similar ways when supported by AI. In this sense, generative AI may not limit the diversity of ideas generated, but it may influence how those ideas are articulated, resulting in greater similarity in linguistic expression.

This finding is particularly relevant given that the deliverable was a handwritten manuscript, meaning the ideas were not copied directly from AI outputs but were manually written by participants. The fact that semantic convergence still occurred despite this manual process suggests that homogenising effects are not limited to direct copy-paste behaviour but may arise through more subtle forms of interaction with AI.

From an educational perspective, this semantic convergence suggests that the language patterns embedded in AI models subtly shape how students articulate ideas, potentially narrowing their range of expression even when conceptual diversity is maintained. While this can enhance clarity and coherence, it may also limit individual expression and stylistic approaches that favour originality and uniqueness. This is particularly important since uniqueness is a defining attribute of creativity, distinguishing genuinely original work from the ordinary (Runco & Jaeger, 2012). Therefore, educators integrating AI into creative tasks should adopt strategies that prompt students to rephrase, critically evaluate, and edit AI-generated language. In writing-related contexts, one approach could be to use AI during initial brainstorming and then set it aside for later development, allowing students to elaborate in their own words and style. Nevertheless, it can be challenging to fully counteract the priming effect once students have been exposed to AI-generated language (Mahowald et al., 2016).

5.4. Student-AI interaction patterns

The analysis of student prompts (Table 8) showed a relatively low engagement, with an average of 6.96 prompts per conversation, 2.31 categories explored, and only eight words per prompt. In addition, the coding results (Table 9) revealed that the *Generic Idea Generation* category was the most frequent, accounting for nearly half of all prompts (48.39 %). In contrast, categories such as *Useful Information, Idea Development, Innovation* and *Opinion* appeared far less frequently. This reliance on broad, open-ended brainstorming suggests a limited use of precise and iterative queries, which are essential for a deeper and more strategic engagement with generative AI. Although time constraints may have contributed to this brevity, the findings suggest that students adopted minimal prompting strategies that yield large volumes of superficially developed ideas, reducing the creative potential of their interactions with AI. Achieving effective co-creativity requires learners to maintain control over their ideas and actively shape the interaction by questioning, iterating, and expanding on AI outputs, rather than passively accepting them (Moruzzi, 2025; Vinchon et al., 2023). However, students' use of AI often lacked these principles of shared agency and reflective engagement, missing opportunities to refine and adapt ideas through genuine collaboration with the system (Lim et al., 2023; Vinchon et al., 2023).

Beyond the formal coding results, interpretive reflections revealed additional patterns in students' interactions with the AI tool. A recurring observation was that, students tended to adopt a passive role, often relying on ChatGPT to generate ideas rather than engaging in independent thinking. This passivity is especially concerning because, as Lim et al. (2023) pointed out, to promote meaningful learning, AI must go beyond functional task completion and promote dialogue, reflection, and emotional engagement.

When learners passively accept AI-generated content, they bypass these deeper processes, undermining the potential of AI as a tool for learning rather than mere production.

These findings underscore that, without proper guidance, AI is unlikely to support meaningful student participation in creative tasks, highlighting the importance of enhancing its pedagogical value through strategies that foster active student engagement. Such strategies should promote reflective and exploratory prompting, encouraging students to question AI's ideas, seek feedback, and experiment with diverse approaches to support intentional and meaningful human-AI collaboration. Building on this need, educators should explicitly teach prompt engineering before assigning AI-based activities, strengthening students' ability to formulate iterative, context-rich, and purposeful prompts (D. Lee & Palmer, 2025). Effective instruction should incorporate concrete examples of successful AI interactions, structured exercises in which students practise refining their queries, and opportunities for students to observe how variations in prompt formulation affect the quality and relevance of results. Educators can also require students to justify their decisions when using AI and reflect on both their prompts and the outcomes produced. These practices can foster students' creative agency in AI-supported tasks, understood as their capacity to make autonomous, intentional, and meaningful contributions rather than passively accepting AI-generated outputs (D. Lee & Palmer, 2025).

Moreover, students frequently used imperative language, such as "*Give me more ideas*" or "*Give me improvements in bus design*", suggesting a transactional perception of AI as an output generator instead of a collaborative partner. They also prioritised quantity over quality, as illustrated by prompts like "*Give me 20 more ideas*" or "*A brainstorm of 50 ideas*". This approach typically involved little follow-up, with students rarely revisiting, evaluating, or improving upon the suggestions they received. This tendency to frame AI in strictly utilitarian terms aligns with Marrone et al. (2022), who found that students often treat AI as a technological tool rather than engage with its creative potential, and with Lim et al. (2023), who observed that students frequently perceived it as unresponsive, which reduced feelings of collaboration and constrained transformative creativity. While emphasising breadth over depth fits brainstorming's aim to generate many ideas, it can result in superficial and less novel contributions, limiting opportunities for more meaningful development. To counter this, students should be encouraged to position AI as a creative partner rather than a source of quick answers. For instance, assigning AI a specific role, such as a domain expert, may prompt students to view it as a knowledgeable collaborator instead of a passive tool (Phoenix & Taylor, 2024). Additionally, prompts beginning with "why" or "how" can stimulate deeper analysis, critical thinking, and more reflective engagement.

Additionally, students often requested innovative ideas without providing context, leading to vague and unfocused AI responses. As noted by B. Chen et al. (2023), broad or ambiguous prompts tend to produce generic outputs, whereas clear and specific instructions are more likely to yield relevant results. Encouraging students to include contextual information and craft precise prompts, such as specifying the type of innovation (e.g., incremental or disruptive), focusing on specific topics (e.g., security or sustainability), requesting the integration of particular technologies, or exploring hypothetical scenarios, can lead to more refined and contextually appropriate suggestions (Knoth et al., 2024; Phoenix & Taylor, 2024).

Finally, despite being familiar with creativity-enhancing methods from prior coursework, students did not incorporate these strategies when interacting with ChatGPT. Instead, they defaulted to a reactive, unstructured mode of seeking ideas. Applying specific ideation methodologies and creativity frameworks could help students realise AI's potential, leading to more diverse, thoroughly developed ideas and fostering a more intentional and effective creative process (Jiang & Luo, 2024; Kim & Maher, 2023; Perez, 2024). For example, students could use the random word association method by prompting ChatGPT to generate a list of unrelated words and then apply the forced connections technique to explore how each term might be meaningfully linked to their project concept.

5.5. Limitations

Although this study provides valuable insights into how generative AI influences ideation processes and creativity, the following limitations should be acknowledged.

As is often the case in research involving AI tools, the findings are sensitive to model updates, and newer systems may yield different interaction patterns and outcomes. Additionally, the study was conducted in a controlled, time-limited experimental setting that may not fully reflect open-ended, self-directed contexts, where students have greater autonomy and more time to refine their ideas. Moreover, the research was conducted with third-year industrial design engineering students from a single institution. To improve the generalisability of the findings, further studies could extend the research to other academic disciplines, levels of expertise, and professional environments.

Regarding the crossover design, while it enabled the comparison of condition order, it may have introduced carryover effects, such as fatigue, that could have subtly influenced the outcomes. Although roles were counterbalanced to mitigate this risk, such effects cannot be entirely ruled out. In addition, the study captures the influence of AI on creativity at a single point in time. Future longitudinal studies could analyse how repeated use of AI influences long-term creative development.

Furthermore, the study employed a mixed between-within subjects ANOVA to assess interaction effects under conditions of non-normality. Although assumptions were carefully examined and complementary non-parametric analyses supported the findings, the results should be interpreted with caution. Finally, since the novelty metric was developed specifically for this study, it would benefit from further validation to strengthen its reliability and applicability in future research.

6. Conclusions

This study deepens the understanding of how generative AI influences creativity by providing empirical insights into undergraduate students' use of ChatGPT during the idea generation process. Using a mixed-methods approach, it highlights strengths and

limitations of AI-assisted ideation.

The results indicate that although AI support led to more semantically similar ideas, it did not reduce the variety of themes explored. On the contrary, students working with AI introduced new topics, maintaining a level of thematic diversity comparable to those who worked without the tool.

While AI did not negatively affect students' fluency, flexibility, or originality scores, the ideas generated with its assistance were more frequently recognised as existing solutions and tended to be less elaborated. These findings call for further investigation and thoughtful pedagogical consideration.

Moreover, the timing of AI use during the ideation process had a significant impact on students' creative performance. Students who began the task with AI produced more flexible and original ideas overall, even in the second task without access to ChatGPT. This suggests that early interaction with AI may support the development of creative thinking strategies that persist beyond the direct use.

Finally, the qualitative analysis of student-AI interactions revealed a frequent reliance on generic and transactional prompting styles, reflecting a passive engagement with the tool. This pattern underscores the importance of enhancing students' AI literacy and creative agency, not only to improve technical proficiency but also to encourage more active and exploratory forms of co-creation with AI that can better support creative thinking.

In sum, although AI can broaden conceptual exploration and support creative thinking, it may also lead to output similarity, reduced novelty, and passivity if not critically guided. As generative tools become increasingly integrated into learning environments, it is essential to equip students with the skills and mindsets necessary to engage with them actively, critically, and creatively. Supporting this kind of engagement is crucial for ensuring that AI serves as a complement to, rather than a constraint on, students' creative development.

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CRediT authorship contribution statement

Rosó Baltà-Salvador: Writing – review & editing, Writing – original draft, Validation, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Enric Brasó-Vives:** Writing – review & editing, Visualization, Validation, Methodology, Investigation, Formal analysis, Conceptualization. **Marta Peña:** Writing – review & editing, Validation, Supervision, Formal analysis.

Declaration of competing interest

The authors declare no competing interests related to this work.

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Data availability

Data will be made available on request.

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